

SEMESTER-III

UMT307: ANALOG AND DIGITAL CIRCUIT DESIGN				
	L	T	P	Cr
	2	1	2	3.5
Course Objective: The aim of this course is to familiarize the student with the analysis and design of basic transistor amplifier circuits, oscillators and wave shaping circuits, basic combinational and sequential circuits in digital domain.				
Syllabus Transistor Biasing and Thermal Stabilization: The Operating Point, Biasing Stability, Self-Biasing or Emitter Bias, Stabilization against Variations in I_{CO} , V_{BE} , and β , Bias Compensation, Thermal Runaway, The metal-oxide-semiconductor FET (MOSFET), The low-frequency common-source and common-drain amplifiers, Biasing FET Amplifiers: Classification of amplifiers, The RC-coupled amplifier, Multistage CE amplifier, Power Amplifiers: Class A, B, AB, Push pull & Class C amplifiers, Comparison of their Efficiencies, Types of distortion. Feedback Amplifiers: The feedback concept, The transfer gain with feedback, General characteristics of negative-feedback amplifiers, Input resistance, Output resistance. Wave shaping Circuits: Sinusoidal Oscillator, The phase-shift oscillator, Resonant-circuit oscillators, The Wien Bridge oscillator, Crystal oscillator, Multi-vibrators (Astable, Mono-stable, Bi-Stable), High pass and low pass filters using R-C Circuits and R-L, R-L-C Circuits & their response to step input, Pulse input, Square input and Ramp Input, introduction to 555 IC and its applications. Binary Codes: Review of special binary codes, Error detection and correction codes. Combinational Circuits: Q. M. Method, Ripple carry adder, BCD adder, Subtractor, Code conversion, Magnitude comparators, Applications of Encoders, Decoders, MUX, DEMUX, Standard ICs and their applications. Using combinational modules to design digital systems Sequential Circuits: Various types of latches and flip-flops and their conversions, Universal Shift Registers, Counters – Ring, Johnson, Design of Counters, Timing issues, Setup and hold times, operating frequency limitations, Standard ICs for their applications IEEE/ANSI standard symbols.				
Laboratory Work Study of Multi-vibrators (Astable, Mono-stable, Bi-stable Multi-vibrator). Hartley and Colpitts Oscillator. RC Phase shift oscillator. Schmitt Trigger. To study standard ICs and their usage, To study latches and Flip-flops, Design of registers and asynchronous/synchronous up/down counters, Study of timing waveforms, Usage of IC tester.				
Course Learning Outcomes (CLOs) The students will be able to: 1. determine operating point and various stability factors of transistor. 2. evaluate the performance parameters of various multistage and power amplifiers. 3. analyse various filters and multi-vibrators circuits. 4. perform Logic Minimization for single/multiple output function(s). 5. design and analyze combinational and sequential circuits.				
Text Books				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

1. Electronic Devices and Circuit Theory, Boylestad, R. L., Pearson 11th ed., 2017
2. Integrated Electronics, Milliman, J. and Halkias, C.C., Tata McGraw Hill, 2007.
3. Digital Design, Mano, M.M. and Clitti M. D., Prentice Hall 3rd ed., 2001.

Reference Books

1. Electronic principles, Malvino, L., Tata McGraw Hill, 1998.
2. 2000 Solved Examples in Electronics, Cathey, J. J., McGraw Hill, 1991.
3. Pulse, Digital and switching waveforms, Milliman, J. & Taub, H., Tata McGraw Hill, 2007.
4. Digital Principles and Design, Givone D. D., Tata McGraw Hill 2nd ed., 2007.
5. Digital Systems: Principles and Applications, Tocci, R.J., Prentice-Hall 10th ed., 2006

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1.	MST	25
2.	EST	40
3.	Sessionals (May include Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	35

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